

Haider Dawood Khan

Lahore, Pakistan

+92 300 8542837 | haiderdawood92@gmail.com | [linkedin.com/in/haider-dawood](https://www.linkedin.com/in/haider-dawood) | github.com/HaiderDawoodKhan

EDUCATION

Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

Bachelor of Science in Computer Science

May 2026

- **cGPA: 3.97**, ranked 4th (Top 2%) in BS CS cohort.
- **Relevant Coursework:** Advanced Topics in Machine Learning[†], Deep Learning[†], Foundations of Generative AI[†], Internet of Things[†], Computer Vision[†], Machine Learning, Data Science, Linear Algebra, Calculus I and II, Probability, Statistics. ([†] Graduate Level Courses)

HONORS AND ACHIEVEMENTS

- Ranked **1st** (2022–23) and **2nd** (2023–24) in LUMS SBASSE (and CS), Batch 2026.
- Recipient of **LUMS Merit Scholarship:** 100% (2023-25) and 50% (2025-26).
- Placed on **LUMS Dean's Honor List** (2022-26).
- IBDP 44/45 — **2nd highest** in Pakistan; TNS Beaconhouse Honor Roll (2020–22).

PUBLICATIONS

PixelPacker: Background-Aware Neural Image Compression for Wildlife Systems

Abdullah Bin Faisal, **Haider Dawood Khan**, Muhammad Fahd Ibrahim Bhatti, Umer Irfan, Ahsan Usman, Muhammad Ali Siddiqi, Naveed Anwar Bhatti (*Under Review at IEEE PerCom 2026*)

RESEARCH EXPERIENCE

Towards Interpretable and Generalizable Audio Deepfake Detection

CSaLT, LUMS

Supervised by Dr. Agha Ali Raza and Dr. Awais Athar

Aug. 2025 – Present

- **Co-first author** of an ongoing project on evaluating domain generalization strategies under **cross-lingual** and **cross-generation** shifts in audio deepfake detection (target: ACL 2026).
- Implemented domain generalization algorithms (**IRM, SAM, ASAM, GDRO, CORAL, MMD, etc**) for training RawNet2 and AASIST models across multiple dataset configurations. Observed competitive baseline for ERM - **>90% accuracy** under language shift but a drop to **~40%** under generation shift on MLAAD, revealing asymmetric generalization.
- Currently evaluating other models and algorithms and developing an interpretability framework integrating model-centric and dataset-centric analyses to explain generalization and learned representations.

Edge-Friendly Neural Image Compression

SysNet & Embedded Systems, LUMS

Supervised by Dr. Naveed Anwar Bhatti and Dr. Muhammad Ali Siddiqi

May 2025 – Oct 2025

- Developed a **hardware-optimized neural image encoder** using depthwise separable convolutions, residual connections, and knowledge distillation from MS-ILLM with quantization-aware training, achieving a **36× lower memory footprint** with comparable image quality.
- Proposed an edge-friendly background-aware compression method eliminating hyperpriors via low-entropy latent modeling, attaining **120× memory reduction** and compression rates as low as **0.001 bpp**.
- Validated on FPGA with **10× higher compression rates than JPEG** at similar perceptual quality; co-first author of a paper submitted to *IEEE PerCom 2026*.

Assessing the Accuracy-Fairness Tradeoff in Language Models

Skidmore University, Remote

Supervised by Dr. Esmā Wali

Nov. 2025 – Present

- Investigating the **accuracy-fairness tradeoff** in multilingual LLMs, focusing on how optimization amplifies gender, racial, and linguistic biases across languages.
- Designing an evaluation framework to quantify **cross-lingual fairness degradation** under performance maximization, addressing gaps in bias metrics and intersectional coverage.
- Building systematic benchmarks and correlation analyses to inform **fairness-preserving optimization** strategies in multilingual NLP.

WORK EXPERIENCE

Motive

USA, Remote

AI/ML Intern (Research and Development)

Jun. 2025 – Sep. 2025

- Utilized SAM for precise leading vehicle pixel segmentation to extensively **evaluate monocular depth estimation models** and transformation strategies for alignment with radar distances and pseudo-distances.
- Curated a **large-scale dataset of 200K+ metric-aligned dense distance map-image pairs** through a hybrid-binning approach utilizing DINOv2 and DepthAnythingV2 to minimize error via radar-guided supervision, **achieving over 20% error reduction** compared to existing approaches.
- Extended and **trained the YOLOX backbone with a depth estimation head** for accurate dense distance prediction and initiated integration of segmentation and depth heads for **one-shot instance segmentation**.

PROJECTS

Inductive Biases and Generalization of Deep Learning Models

- Fine-tuned **ResNet-50** and **ViT-B/16** on STL-10 and PACS datasets to compare color, texture, and shape biases, revealing that CNNs exhibit stronger texture bias while ViTs capture human-aligned shape representations.
- Trained **VAE** and **DCGAN** models on CIFAR-10 to study the **fidelity–diversity trade-off**, showing VAEs' mode-covering and GANs' mode-seeking tendencies, quantified via FID (132 vs. 30).
- Evaluated **CLIP's zero-shot generalization** across CIFAR-10 and PACS, achieving 88.8% accuracy and demonstrating superior domain robustness on sketches and cartoons.

Model Compression: Pruning, Quantization & Distillation

- Performed structured (channel) and unstructured (magnitude) pruning on VGG-11 fine-tuned on CIFAR-10 using layer-wise sensitivity and a prune-evaluate-finetune loop; unstructured sparsity $\approx 70\%$ recovered/surpassed baseline after finetuning (93.04% vs 92.95%) without MAC savings, whereas structured pruning cut MACs by **68.7%** and latency by **65%** at 91.84% accuracy.
- Built PTQ/QAT pipelines across different bit ranges and showed **INT8 QAT** regained near-FP32 performance within one epoch while **INT4** remained capacity-limited; a Hessian-aware mixed-precision scheme kept early/last layers high precision and matched FP32 accuracy (70.9%) at $\sim 4\times$ smaller size.
- Compared distillation methods (Logit Matching, DKD, Hints, CRD) for VGG-16 $\rightarrow$ VGG-11 on CIFAR-100, with **Hints** yielding the best student accuracy (67.73%, $\approx$ teacher 67.84%) and highest Grad-CAM similarity (0.73), while **Logit Matching** achieved the lowest KL (0.1178), evidencing an accuracy–fidelity trade-off.

Federated Learning under Statistical Heterogeneity

- FedAvg**, showed IID gains from larger local work (K: 1 $\rightarrow$ 20 improving accuracy 90.35% $\rightarrow$ 96.81% but raising drift) and that client sampling is safe on IID (minimal accuracy drop), yet harmful on non-IID.
- On non-IID, **SCAFFOLD** achieved the best accuracy **91.48%** and the lowest drift **0.0685** (vs FedAvg 0.8736 / 0.2279) at $2\times$ communication (131.3M vs 65.6M bytes); **FedProx** matched FedAvg for small μ and reduced drift by $\sim 15\text{--}18\%$ at $\mu=0.5$ with a small accuracy drop.
- FedSAM** improved non-IID accuracy to **0.8980** at $\rho=0.1$ (monotonic gains with ρ) while increasing drift to **0.3220** and adding per-round compute; IID gains were modest but consistent.

Automated Number Plate Detection & Logging System

- Fine-tuned a **YOLO**-based detector to localize license plates in road images; exported a lightweight inference routine for fast per-image detection.
- Built a character recognition pipeline using **morphological preprocessing + contour filtering** for segmentation and a custom **36-class CNN** (A–Z, 0–9) in PyTorch; achieved **99% accuracy** on the validation set.
- Developed a **Streamlit** app to **search plates by text** and instantly view the corresponding vehicle image and log metadata for quick lookups.

Training Diffusion and Conditional Diffusion Models

- Trained a DDPM on MNIST using a **cosine noise schedule** and U-Net denoiser with **sinusoidal timestep embeddings** and trained the model using **MSE** to optimize the reconstruction objective.
- Added **class-conditional diffusion** via learnable **class embedding** layers to synthesize on-class samples.
- Integrated **CLIP-guided loss** for **prompt-conditioned** generation, steering samples toward prompt semantics.

SmolLM Variant with LoRA and DPO

- Implemented a lightweight decoder-only Transformer with **RMSNorm**, **RoPE**, **GQA** attention, and **SwiGLU** MLP; modular backbone, decoder, and LM head.

- Integrated **LoRA** adapters into attention and FFN blocks, cutting **~80%** of trainable parameters and VRAM during fine-tuning while preserving task performance.
- Implemented the **Direct Preference Optimization** objective with a frozen reference model, pairwise preference dataloader, and β temperature scheduling for better alignment.
- Improved chosen-response selection accuracy by **16%** on held-out preference pairs, achieving better alignment without RLHF.

Brandenburg Gate 3D Reconstruction

- Performed feature detection with **SIFT** and feature matching with **FLANN**, using **Lowe's ratio test** to retain high-quality correspondences.
- Estimated camera geometry by computing the **Fundamental Matrix**, deriving the **Essential Matrix** from intrinsics, decomposing it, and **triangulating to obtain a sparse 3D point cloud** and camera poses.
- Applied **Multi-View Stereo** to densify the reconstruction, producing a dense 3D point cloud of the landmark.

Edge AI Human Activity Recognition

- Trained a TF model on 20 Hz IMU windows to detect activities such as walking, jogging, typing, writing, cycling, and standing and performed post-training MP quantization and exported header file for MCU deployment.
- Deployed on **ESP32** with **TFLite Micro**: buffered 40-sample MPU-6050 windows, invoked the interpreter on-device, and applied majority voting over the last 60 s to stabilize predictions.
- Built a lightweight web page/backend to display current activity and approximate step count via per-user calibration with periodic updates.

LEADERSHIP

Senior Lab Lead

LUMS

Center for Speech and Language Technologies (CSaLT) — Led by Dr. Agha Ali Raza

May 2025 – Present

- Selected as **one of four Senior Lab Leads** from a pool of over **150 candidates** through a rigorous merit-based process recognizing outstanding academic and research performance.
- Supervising **31 undergraduate researchers** undertaking Senior and Directed Research Projects, ensuring technical rigor, timely deliverables, and effective mentorship.
- **Coordinating all ongoing lab projects**, including student-led and externally funded research, by maintaining structured progress tracking, documentation, and milestone reviews.
- Managing lab operations, equipment logistics, and internal communication systems.

TEACHING EXPERIENCE

CS 331: Foundations of AI and Machine Learning

Dr. Agha Ali Raza

Head Teaching Assistant

Aug 2025 – Present

- Led a team of **12 TAs** and supported **300+** students; handled course logistics, assisted in course design, grading workflows, and course communication.
- Designed NN assignments (forward/backward from scratch, initialization, PyTorch) and paper-based quizzes on Naive Bayes and ensemble methods.
- Held office hours; **automated grading pipelines** and managed exam logistics/invigilation.

AI 500: Foundations of AI (Graduate Course)

Dr. Murtaza Taj

Teaching Assistant

Aug 2025 – Present

- One of **four TAs** for **110+** MS students; delivered tutorials on KDE, SVMs, and search.
- Authored and graded problem sets, assignments, and weekly quizzes (causal inference, EDA, GMMs, clustering).
- Developed rubrics and conducted focused office hours to strengthen conceptual understanding.

CS 334: Principles and Techniques of Data Science

Dr. Ihsan Ayub Qazi

Teaching Assistant

Aug 2024 – Dec 2024

- Only junior among five TAs for **160+** students; conducted tutorials on supervised learning, EDA, statistics, etc.
- Created Python assignments (pandas, decision trees, random forests) and mentored course projects.
- Supported exam preparation and grading with consistent feedback standards.

Other Courses (Teaching Assistant)

CS 200: Introduction to Programming & CS 100: Computational Problem Solving

Aug 2024 – May 2025

- Supported **300+** students across two introductory courses; led labs/recitations on C/C++ fundamentals (strings, loops, recursion, STL).
- Built hands-on problem-solving tutorials and assisted with assignment/lab grading.
- Managed course tools (Slack), coordinated viva logistics, and provided targeted one-on-one mentorship.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, SQL, Javascript, Typescript, Bash, Haskell

Frameworks/Libraries: React, Node.js/Express, pandas, NumPy, Matplotlib, Seaborn, PyTorch, TensorFlow, TFLite, MMCV, Flask, FastAPI, scikit-learn, Socket.io, cv2, LangChain, HayStack, Selenium

Cloud/Tools: AWS EC2/S3/Lambda, Git/GitHub, Linux, Jupyter, Postman, MongoDB, VSCode, Google Colab, Kaggle, Docker